



Fig. 6: STK500 window for data downloading into microcontroller

click 'Stop Watch' and reset the stop watch. Continue debugging by pressing F11 key and note the time when it is cleared again and finally set. This

should be approximately one second. After this, code simulation is complete. You have successfully coded and simulated your AVR program.

In the following section, you will practically download your code to ATmega8515 flash memory and see things happening.

Circuit description

Fig. 5 shows the circuit of the development board for ATmega8515. The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 9V, 500mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7805 (IC1). Capacitor C2 bypasses the ripples present in regulated supply. LED1 acts as the power indicator and resistor R2 limits

PARTS LIST

Semiconductors:

- IC1 - ATmega8515 AVR microcontroller
- IC2 - MAX232 RS-232 driver
- IC3 - 7805, 5V regulator
- D1-D4 - 1N4007 rectifier diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, 5% carbon):

- R1 - 10-kilo-ohm
- R2 - 330-ohm

Capacitors:

- C1 - 1000µF, 25V electrolytic
- C2 - 0.1µF ceramic disk
- C3 - 3.3µF, 16V electrolytic
- C4-C9 - 1µF, 16V electrolytic
- C10, C11 - 22pF ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 9V, 500mA secondary transformer
- S1 - Push-to-on tactile switch
- X_{1AL} - 16MHz ceramic crystal
- 9-pin D-type male connector
- 3-pin berg strip (male)
- 6-pin ISP male connector
- 3x8-pin SIP male connector

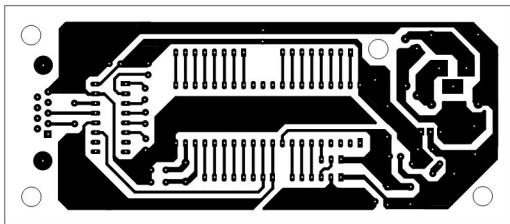


Fig. 7: An actual-size, single-side PCB for the ATmega8515 development board

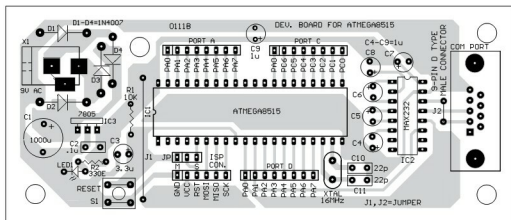


Fig. 8: Component layout for the PCB

the current through LED1.

Atmega8515 is the heart of the circuit. It features 8kB Flash memory with read-while-write capabilities, 512-byte EEPROM, 512-byte SRAM, 35 general-purpose input/output lines, 32 general-purpose working registers, two flexible timers/counters with compare modes, internal and external interrupts, a serial programmable USART, a programmable watchdog timer with internal oscillator and SPI serial port.

Switch S1 is used for manual reset. Port pins PB5 (MOSI), PB6 (MISO) and PB7 (SCK) are used for in-system programming (ISP). IC MAX232 (IC3) is used for serial communication. TXD and RXD pins of the microcontroller are used for serial communication with the help of COM port.

An actual-size, single-